

# Oestrogen levels in serum and urine of premenopausal women eating low and high amounts of meat.

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**OBJECTIVE:** Based on the hypothesis that high-meat diets may increase breast cancer risk through hormonal pathways, the present analysis compared oestrogens in serum and urine by meat-eating status. **DESIGN:** Intervention with repeated measures. **SETTING:** Two randomized soya trials (BEAN1 and BEAN2) among premenopausal healthy women. **SUBJECTS:** BEAN1 participants completed seven unannounced 24 h dietary recalls and donated five blood and urine samples over 2 years. BEAN2 women provided seven recalls and three samples over 13 months. Serum samples were analysed for oestrone (E $\alpha$ ) and oestradiol (E $\beta$ ) using RIA. Nine oestrogen metabolites were measured in urine by LC-MS. Semi-vegetarians included women who reported consuming <30 g of red meat, poultry and fish daily, and pescatarians those who reported consuming <20 g of meat/poultry but >10 g of fish daily. All other women were classified as non-vegetarians. We applied mixed models to compute least-square means by vegetarian status adjusted for potential confounders. **RESULTS:** The mean age of the 272 participants was 41.9 (SD 4.5) years. Serum E $\alpha$  (85 v. 100 pg/ml,  $P = 0.04$ ) and E $\beta$  (140 v. 154 pg/ml,  $P = 0.04$ ) levels were lower in the thirty-seven semi-vegetarians than in the 235 non-vegetarians. The sum of the nine urinary oestrogen metabolites (183 v. 200 pmol/mg creatinine,  $P = 0.27$ ) and the proportions of individual oestrogens and pathways did not differ by meat-eating status. Restricting the models to the samples collected during the luteal phase strengthened the associations. **CONCLUSIONS:** Given the limitations of the study, the lower levels of serum oestrogens in semi-vegetarians than non-vegetarians need confirmation in larger populations.